Should You Get Screened for Diabetes?
A Deep Dive Into Variables Connected With Diabetes

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1. Introduction

Diabetes Mellitus, commonly referred to as diabetes, is a chronic metabolic disorder characterized by elevated blood glucose levels (*Diabetes*, n.d.). Over recent decades, the global prevalence of diabetes has risen dramatically, positioning it as a significant public health concern worldwide (*IDF Diabetes Atlas*, n.d.).

As of 2022, it is estimated that over 800 million adults globally are living with diabetes, indicating a substantial increase from previous years (Reuters, 2024). The number of individuals with diabetes is projected to rise to 643 million by 2030 and 783 million by 2045 (*Facts & figures*, n.d.).

In regions such as North America and Europe, diabetes prevalence is notably high. For instance, the United States has approximately 12.5% of its adult population diagnosed with diabetes. Over 75% of adults with diabetes reside in low- and middle-income countries, where access to healthcare and treatment options may be limited. Diabetes is associated with various complications, including cardiovascular diseases, kidney failure, nerve damage, and vision loss. In 2021, diabetes was responsible for approximately 6.7 million deaths globally, equating to one death every five seconds. The escalating prevalence of diabetes imposes a significant economic burden on healthcare systems. Global health expenditure on diabetes is estimated to be around USD 760 billion annually, accounting for 10% of total health expenditure (Bawden, 2024).

In conclusion, diabetes poses a growing global health challenge and this project aims to predict a model which is able to describe the variables and the risk of occurrence. To achieve this, a dataset from Kaggle, a platform for sharing datasets online, is utilized. The dataset is a cleaned and consolidated version and contains around 250.000 individuals with or without diabetes. For each patient different socio-economic, health related and behavioral features were collected (*Diabetes Health Indicators Dataset*, n.d.).

1.1 Limitations and Potential Issues

This dataset is comprehensive but comes with some limitations. Firstly, the data does not include genetic predisposition, medication use, or other potential aspects that may impact diabetes risk. Additionally, lifestyle habits such as smoking, alcohol consumption, and physical activity are self-reported, which introduces the possibility of response bias, which could alter the conclusions of the analysis. The pre-processing applied

Moreover, the age feature in the dataset is encoded as categorical values ranging from 1 to 13, rather than representing actual numerical ages. This encoding likely corresponds to age groups, where each value represents a specific range (e.g., 1 = 18-24, 2 = 25-29, etc.). This limits precise statistical analysis and makes it difficult to calculate meaningful averages or distributions. Similarly, the standard deviation of 3.05 does not provide a true measure of age variability.

2. Responsible Data Guiding Principles

D’Ignazio and Klein developed guidelines for feminist data visualization (D’Ignazio & Klein, n.d.). While focusing on a feminist perspective, the six principles are good guidelines for a responsible data analytics project.

2.1 Rethink Binaries
The extent to which we are able to respect non-binary people is limited by the pre-processing applied to our dataset. During the pre-processing that was applied to the dataset before being published on Kaggle, persons with non-binary genders were removed. This introduces a representation bias into our dataset. Furthermore, in the survey, the subjects had to answer in predefined categories for most questions. While this makes for easier analysis and pre-processing, it also diminishes the chance to collect the diverse contexts that most people live in. These issues are further investigated in the paragraphs about potential biases.

2.2 Embrace Pluralism
Our team consists of five students from European countries. While being from different countries means that we have diverse cultural backgrounds, all of us have access to a good education and a good healthcare system. This may lead to our team being less able to incorporate the views and ideas of subpopulations into which we have no direct insight. We are going into more depth on this topic in our group positionality and reflexivity statement.

2.3 Examine Power and Aspire to Empowerment
Since our use case is the determination of a person's risk of being diabetic, feedback on usability, intuitivity and design choices are crucial for high acceptance of the predictive model. This feedback loop could be implemented through a rating mechanism or anonymized surveys. Especially when handling healthcare data, factors indicating a higher risk of having an illness can lead to discrimination or stigmatization. Therefore, we try to be as transparent as possible in the functioning of our utilized model. More information about fairness in this data analytics project can be found in the respective paragraph.

2.4 Consider Context
Especially when investigating research questions about illnesses that can be caused by behavioral practices, considering the context of patients is important. Diabetes can be influenced by unhealthy practices, such as not getting enough physical activity or eating too unhealthily. However, having a low income may not allow the financial freedom necessary to buy healthier grocery products. Furthermore, a stressful job limits the time available to work out. In these cases, the context of people limits their ability to influence the factors that are important for reducing the risk of becoming diabetic.

2.5 Legitimize Embodiment and Affect
Design choices can have a large impact on how easily key messages are communicated to the reader. We strive to make our visualizations as simple as possible while still maintaining the complexity necessary to convey key takeaways. To enhance the affective experience of the reader, we will create all plots using the same color palette. This ensures a coherent user experience across visualizations. Because we mostly show histograms and confusion matrices, we believe that one-sided or stigmatizing visualizations are not a huge issue in our specific case. That being said, we still paid attention to visualize in a neutral way.

2.6 Make Labor Visible
The dataset used was collected through The Behavioral Risk Factor Surveillance System (BRFSS) (*Behavioral Risk Factor Surveillance System*, n.d.) in a telephone survey. The interviewers must be given credit for the individual and time intensive data collection process. Furthermore, Alex Teboul (https://www.kaggle.com/alexteboul), a U.S. based Analyst and Programmer did all the pre-processing and published the dataset on Kaggle where it first came to our attention. Within

our team, we tried to distribute the workload as fairly and evenly as possible. Therefore, we were all equally responsible for the success of this data analytics project.

3. Privacy, Positionality, and Bias I

3.1 Reflection on the Privacy Impacts

The diabetes project raises several privacy concerns, particularly regarding the collection, storage, and use of sensitive health and socio-economic data. Given that the dataset includes variables such as age, income, education, and health conditions, there is a risk of re-identification, even if the data has been anonymized. Individuals' health status is highly sensitive, and improper handling could lead to privacy breaches, discrimination, or misuse by third parties such as insurance companies or employers (*Patient survey shows unresolved tension over health data privacy*, 2022).

Privacy, as discussed in the lecture material, can be understood from different perspectives: confidentiality, control, and ethical considerations. Confidentiality ensures that personal information remains protected, requiring strict access controls and encryption measures. Control refers to individuals' rights over their data, aligning with GDPR principles such as informed consent, the right to be forgotten, and data accuracy (Wolford, n.d.). Ethical considerations highlight the responsibility to safeguard privacy, even when data collection is justified for public health research (Murdoch, 2021).

Another concern is the presence of proxy variables that could indirectly reveal protected characteristics. For example, income and education levels may serve as proxies for race or socio-economic status, potentially leading to unintended biases in the model's predictions (Prince & Schwarcz, 2020). Moreover, self-reported data on health behaviors, such as smoking and alcohol consumption, introduces risks of bias and inaccuracies, further complicating the responsible use of this data (Frost, n.d.).

To reduce privacy risks, implementing strict data governance policies, ensuring transparency in data usage, and applying privacy-preserving techniques like differential privacy can enhance ethical data practices. Additionally, continuous assessment of bias and fairness metrics should be integrated to minimize harm and ensure that the model does not disproportionately impact vulnerable people. Balancing predictive accuracy with privacy considerations is essential for responsible data analytics (Price & Cohen, 2019).

3.2 Group Positionality and Reflexivity Statement

As a group conducting a data analytics project on diabetes risk prediction, we recognize that our backgrounds, experiences, and perspectives influence our approach to data interpretation, analysis, and ethical considerations. We come from five different European countries, all with easy access to healthcare, so we might take it for granted that medical care is available to everyone. Our dataset is based on US data, where access depends more on insurance and income. This difference means that we need to be careful not to interpret the results through a European perspective and consider how the healthcare in the US affect the data.
Our dataset includes factors related to socio-economic status and health that could be affected by inequalities in society. We understand that things like access to healthcare, education, and income distribution differ among different groups, and we need to carefully consider these factors to avoid reinforcing any existing biases. As a group, we strive to maintain objectivity, but

we acknowledge that our choices are influenced subjectively which may impact different demographic groups differently.

Our understanding of diabetes, healthcare systems, and even what "risk" means is shaped by where we come from and what we've experienced. Some of us may see diabetes as a primarily lifestyle-related issue, while others may view it as deeply connected to genetics, access to healthcare, or other factors that people don't have control over. We recognize that we might subconsciously approach the analysis in ways that align with our personal views or experiences. While we strive to be objective, we know that our perspectives shape the way we frame questions, interpret results, and even define fairness. By respecting these influences, we hope to challenge our own biases and create a more balanced and thoughtful analysis. Having open discussions continuously within our group can help give different perspectives and help ensure we challenge our own biases.

3.3 Bias Analysis

When creating equitable and successful prediction models, it is important to address the types of bias that were identified during the study of the diabetic health indicators dataset.
Firstly, with over 141,000 male records compared to 111,000 female records, the dataset contains more male entries than female entries. As a result, models trained with this data may perform better for men than for women. Additionally, even if most people in both categories do not have diabetes, women are somewhat more likely than men to have diabetes, which might inject bias into the model's performance if left unchecked.

Moreover, when historical bias is examined, a certain age-related pattern becomes apparent. The prevalence of diabetes rises sharply with age. While diabetes is far more common in later age groups (45–54 and above), younger age groups (18–24) are overwhelmingly categorized as non-diabetic. While this represents current patterns, it might also record past disparities in access to healthcare or health-related behaviors that are encoded in the data and that models could pick up on and magnify.

In conclusion, the dataset has substantial representation and historical biases even if it offers insightful information about diabetes and the health indicators linked to it. Based on these results, one must bear in mind that predictive models created using this data run the danger of maintaining or escalating already-existing gaps in healthcare outcomes in the absence of such mitigation.

3.4 Historical Biases

Historical biases happen when a model or dataset perpetuates a negative stereotype that already exists in society. Historical bias may have an impact on a number of dataset characteristics in the diabetes study, especially when it comes to socioeconomic and healthcare disparities.

One primary concern is that the dataset may involuntarily perpetuate current socioeconomic inequities. For instance, traditionally, access to healthcare has been more difficult for those from lower-income families, which might lead to underdiagnosis or treatment delays. These disparities can lead to unequal representation of health outcomes across different income groups. In fact, according to the research, 14.82% of the data reflects people with low incomes (income levels 1-2), 32.52% belong to the medium-income group, and 52.67% are high-income (income levels 7-8). Higher-income people are overrepresented in this distribution, which might add to the model's biases, especially when it comes to forecasting diabetes and other health concerns. Lower-income groups, who face greater health inequities, may be underrepresented in the dataset, leading to poorer prediction accuracy for these populations.

Additionally, the dataset may also contain historical biases due to demographic features such as age, education, and sex. Because they have less access to healthcare services and health information, those with lower levels of education are more likely to have poorer health outcomes, which matches the conclusions of the carried analysis.

Furthermore, historical biases in how the medical community has treated particular populations may also be reflected in the dataset. However, as has already been mentioned, this dataset has been anonymized, which makes it mostly impossible to check if this bias is being maintained with the used data. It would be valuable in the future to be able to access this type of data, as it would allow to detect certain historical bias linked with discrimination and allow professionals to avoid it in a more efficient way.

Finally, in aiming to lessen these biases, one must ensure future datasets are inclusive of all demographic groups and that they represent the population accordingly. By using balanced sampling strategies and adding more data sources that address the inequities present in the current healthcare systems, this can be accomplished.

3.5 Representation Biases

Representation bias occurs when a dataset does not accurately reflect the population it aims to model, often leading to unfair or skewed outcomes in predictive analytics. In this project, potential representation biases stem from the demographic composition of the dataset and the way different groups are included or excluded.

One major concern is that the dataset, despite being extensive, may not proportionally represent all demographic groups. Certain populations, such as individuals with lower digital literacy, those without stable housing, or non-English speakers, may be underrepresented due to barriers in data collection methods. Surveys and digital health records often rely on self-reported data or require access to healthcare facilities, which some marginalized groups may not frequently visit.

Additionally, individuals from lower-income backgrounds or rural areas may have less engagement with healthcare systems, resulting in missing or incomplete data. As a result, the model might fail to accurately assess diabetes risk for these groups, leading to disparities in prediction accuracy. If the dataset lacks sufficient representation from these populations, the model may generalize poorly when applied to real-world scenarios, potentially reinforcing existing healthcare inequities (Gallon, 2024).

Furthermore, self-reported variables such as physical activity, diet, and general health perceptions can introduce representation bias. Cultural differences and personal interpretations of health-related questions may lead to inconsistencies in responses. For example, individuals from certain cultural backgrounds might underreport alcohol consumption or overstate their physical activity levels, skewing the relationships between these behaviors and diabetes risk (Spitzer & Weber, 2019).

To address representation bias, a more balanced dataset should be used, ensuring that all relevant demographic groups are adequately represented. This could involve stratified sampling techniques or supplementing the dataset with additional sources that include underrepresented populations (Hasanzadeh et al., 2025).

3.6 Measurement Biases

Measurement bias happens when the way data is collected, recorded, or processed leads to systematic inaccuracies, affecting the reliability of a predictive model. In the diabetes project, several factors contribute to potential measurement bias, particularly in self-reported health behaviors, categorical encoding, and standardized medical metrics.

One source of measurement bias is the reliance on self-reported data for variables such as smoking habits, alcohol consumption, physical activity, and general health perception. Respondents may misreport their behaviors due to recall errors, social desirability bias, or cultural norms. For example, individuals may underreport alcohol consumption or overstate their physical activity levels due to stigma or personal bias. This leads to inconsistencies in the dataset, potentially weakening correlations between lifestyle factors and diabetes risk (Li et al., 2023).

Furthermore, standard medical metrics such as BMI and blood pressure may not fully capture individual health risks. BMI, for example, does not distinguish between muscle and fat mass, potentially misclassifying athletes as overweight or obese. It also does not consider fat distribution, which is a critical factor in metabolic health, as visceral fat poses a greater risk for diabetes and cardiovascular diseases than subcutaneous fat (Muscogiuri et al., 2023). Similarly, blood pressure measurements can fluctuate due to temporary stress, hydration levels, or physical activity, leading to potential misclassification of an individual's long-term health risks (Tomitani et al., 2023). Such limitations in measurement can result in biased model predictions and misinterpretations of diabetes risk.

To mitigate measurement bias, improving data collection methods, validating self-reported data with objective measurements, and using more precise numerical encoding for categorical variables would enhance the model's reliability. Recognizing and addressing these biases is essential to ensure accurate, fair, and meaningful insights from the analysis.

4. Descriptive Analysis

This descriptive analysis explores a dataset related to diabetes and associated health factors. The dataset consists of 253,680 observations and 22 features, including key health indicators such as blood pressure levels (HighBP), cholesterol levels (HighChol), body mass index (BMI), smoking habits (Smoker), and physical activity (PhysActivity). It also contains demographic information such as age, sex, education level, and income, along with health status indicators like general health (GenHlth), mental health (MentHlth), and physical health (PhysHlth).

Using descriptive statistics, this study intends to highlight patterns that may offer insights into diabetes risk factors and overall health conditions among various population groups. The main goal of this analysis is to give an overview of the dataset by looking at the distribution of key variables, identifying trends, and detecting potential correlations among health factors. The prevalence of diabetes (Diabetes_012) and its relationship with lifestyle habits and medical conditions will receive particular attention.

4.1 Initial Observations and Main Covered Areas

To begin with, it is crucial to analyze the data types present in this dataset and what exactly does this data cover. The analysis shows that all columns consist of float64 values, meaning that categorical variables are encoded numerically. (A deeper analysis on each of the features is available in the Feature table located in the appendix)

Additionally, it is key to highlight the absence of missing values, which preliminarily suggests decent quality data. As the analysis also shows, it is possible to identify 4 different aspects covered by the data, which are diabetes status, health indicators, demographics, and lifestyle habits.

4.2 Population Analysis

The case study is from the Behavioral Risk Factor Surveillance System report of 2015. The study is collaborative project between all of the states in the US, participating US territories, and the Centers for Disease Control and Prevention. In conducting the BRFSS landline telephone survey, interviewers collect data from a randomly selected adult in a household. In conducting the cellular telephone version of the BRFSS questionnaire, interviewers collected data from adults who participated by using a cellular telephone and resided in a private residence or college housing.

The sample population is composed of both genders. The age spans from 18 up to 80 years old and over, the average age group falls in the category 50-59 years old. Over 40% of the population sampled has completed four years of college or more, while the rest of the set is mainly of college and high school graduate individuals. About 36% of the sample belongs to the 75.000 class of income.

The survey, conducted across a population from 50 countries, includes several features that may be subject to bias. These include self-reported variables such as difficulty walking (a boolean variable, with 16.8% of respondents indicating difficulty), as well as physical, mental, and general health status over the past 30 days. Other potentially biased features include heavy alcohol consumption and whether individuals needed to see a doctor but couldn't afford it. Since the data is collected through self-reported surveys these variables may be affected by recall bias, social desirability, and cultural differences in reporting.

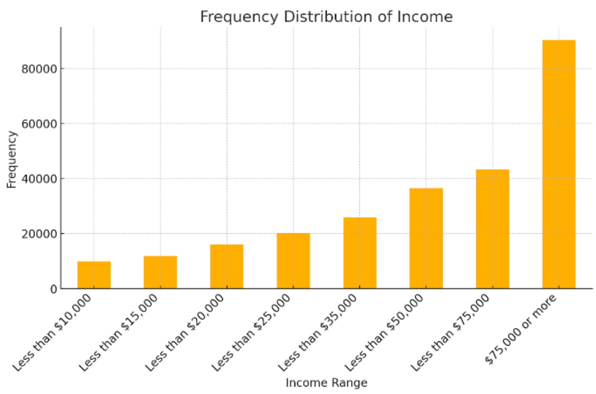

Figure 2 Distribution of Income

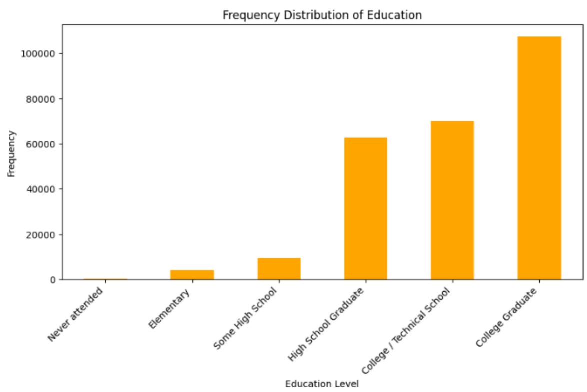

Figure 1 Distribution of Education

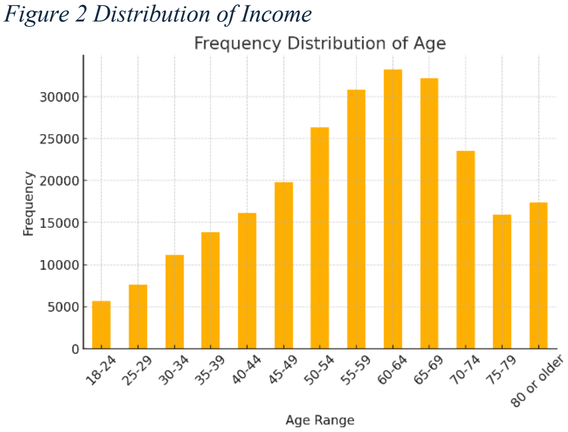

Figure 4 Distribution of Age

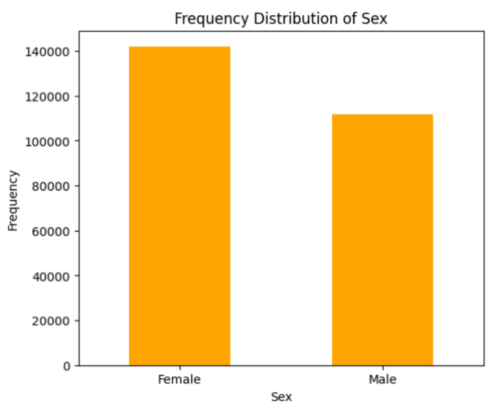

Figure 3 Distribution of Sex

4.3 Descriptive Statistics and Distribution Analysis

The dataset reveals that approximately 15.7% of individuals are classified as either prediabetic (1.83%) or diabetic (13.9%). High blood pressure is observed in 42.9% of individuals, while 42.4% have high cholesterol. About 9.4% report a history of heart disease or attack.

Lifestyle factors indicate that 44.3% of individuals are smokers, 75.6% engage in physical activity, and only 5.61% consume alcohol regularly.

Self-reported general health scores range from 1 (excellent) to 5 (poor), with an average rating of 2.5, suggesting that most individuals perceive their health as fair.

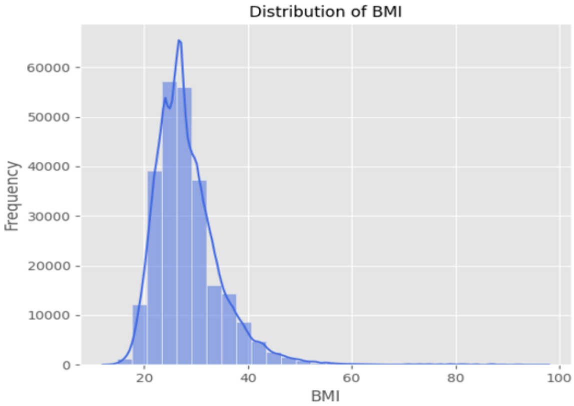

Figure 6 Distribution of BMI

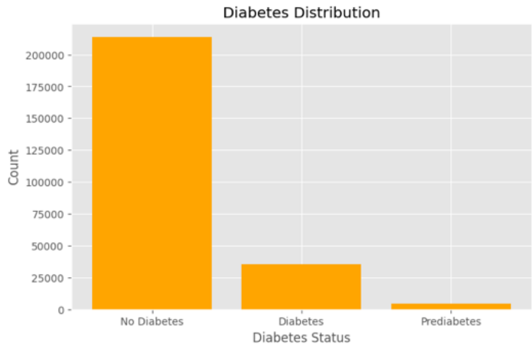

Figure 5 Distribution of Diabetes

Mental and physical health indicators show that, on average, individuals report approximately 3.2 days of poor mental health and 4.2 days of poor physical health per month.

The histogram of the BMI feature shows a skew distribution, with some outliers, which requires a further analysis of the distribution of data points in that dimension. Since the BMI is, from literature, one of the main causes of the increase of risk in diabetes, it is relevant to analyze the significance of the upper outliers and whether they should be considered or not in the predictive model. The boxplot of “BMI Distribution” shows the relevance of the upper outliers, while the lower outliers are far less present. The identification of the outliers has been performed by considering the data points external to the whiskers shown in the plot, to better understand their intrinsic value. The analysis counts an equivalent of 9820 upper and 27 lower outliers, with a percentage of 34.27 % and 7,4 % of people with diabetes in the upper and lower outliers respectively. The total amount of outliers covers 3.93 % of the entire dataset, so it may be reasonable to avoid considering them, to train a more accurate predictive model. However, since in some countries the percentage of people having a BMI greater than 40, considered as “severe obesity” threshold, is around 10 %, as in USA, a model will be trained taking them into account. This choice has been motivated also by the fact that the model did not change in accuracy relevantly with outliers, so it has been decided to train a more inclusive model, to consider extreme cases and minorities as well.

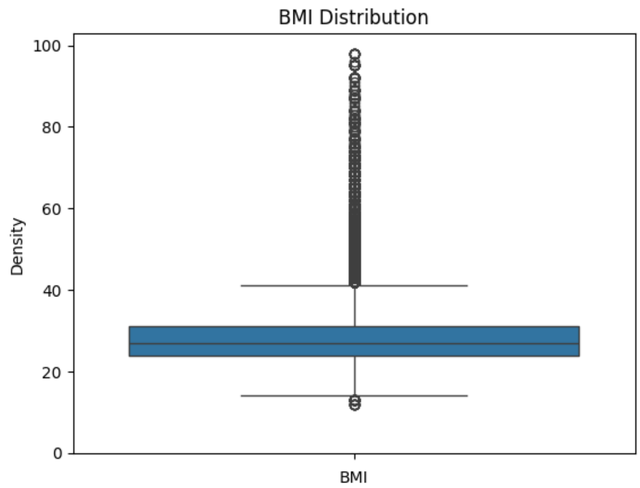

Figure 7 BMI Distribution

The dataset has been analyzed for its significance of the most relevant features, which will be further explored with a clustering, based on these features. BMI, High blood pressure, Physical activity, Fruits and Income are tested. Since BMI and Income are continuous variables, a t-test is performed, to evaluate if the differences with respect to diabetes and non-diabetes individuals are meaningful. The results show p-values of approximately 0 for both the features. A chi-squared test is, instead, performed for the boolean variables of High blood pressure, Physical activity and fruits. P-values are approximately 0 for all the features mentioned, confirming the meaningfulness of the features between diabetes and non-diabetes individuals.

5. Diagnostic Analysis

Firstly, this Diagnostic Analysis aims to understand the correlations between the features and diabetes, but also between the features themselves, so that patterns and clusters are identified. Then, a focus is made on the influence of features on diabetes with the help of regression tools.

5.1 Correlation Analysis

A correlation analysis has been conducted to identify relationships between various features and their association with diabetes.

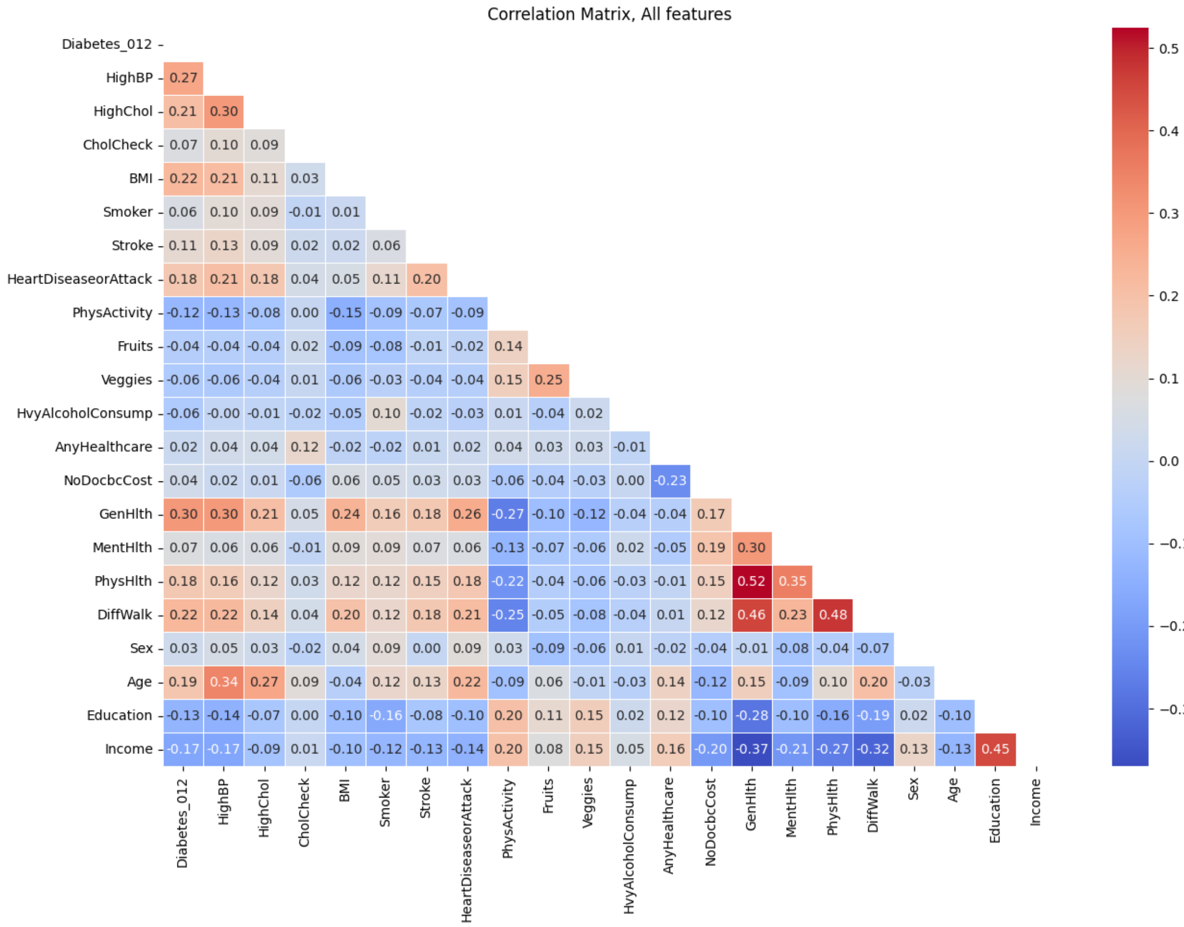

Figure 8 Correlation Matrix for all Features

The results show in the first column that general health rating (0.30), high blood pressure (0.27), BMI (0.22), difficulty walking (0.22), high cholesterol (0.21), and age (0.18) have the strongest positive correlations with diabetes. This suggests that individuals with these risk factors are more likely to be diabetic.

Conversely, higher income (-0.17), higher education (-0.13) and physical activity (-0.12) are associated with a lower prevalence of diabetes.

A diet rich in fruits (-0.04) and vegetables (-0.06), smoking (0.06), alcohol consumption (0.06), sex (0.03) and having healthcare coverage (0.02) are poorly correlated with diabetes.

Moreover, the rest of the matrix highlights that other features are highly correlated. Socio-economic, health and behaviors data correlations are studied to identify if some of these features are redundant in these categories.

The socio-economic correlation matrix demonstrates that education and income are highly correlated (0.45) and that people that have not been able to see a doctor because of cost are more likely to be the ones with low income (-0.20), low education (-0.10) and that don't have any health care coverage (-0.23). When health care coverage and the financial ability to see a doctor don't really influence the likelihood of being diabetics, it can be identified that more privileged people with high income and education level are less subject to diabetes.

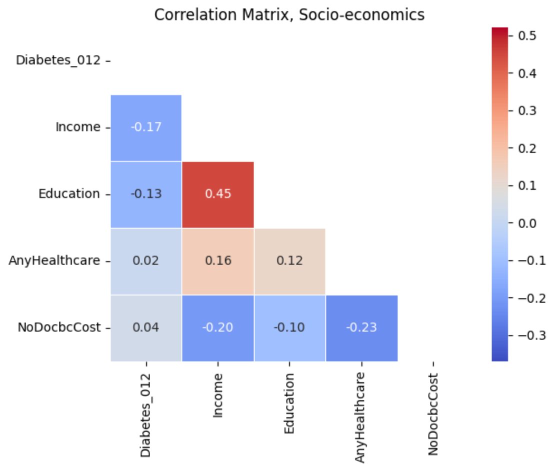

Figure 9 Correlation Matrix for socio-economic features

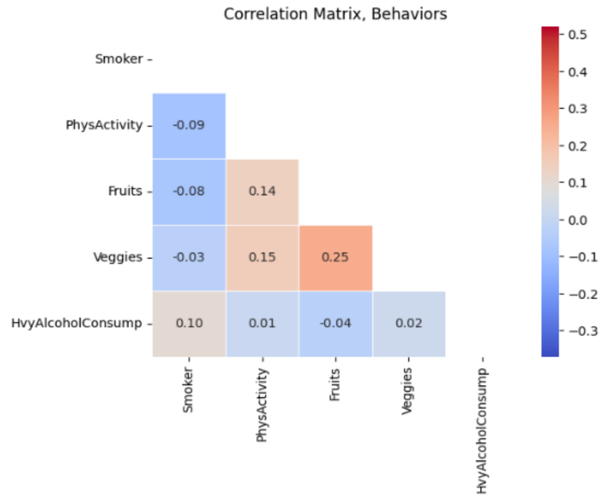

Figure 10 Correlation Matrix for behavioral features

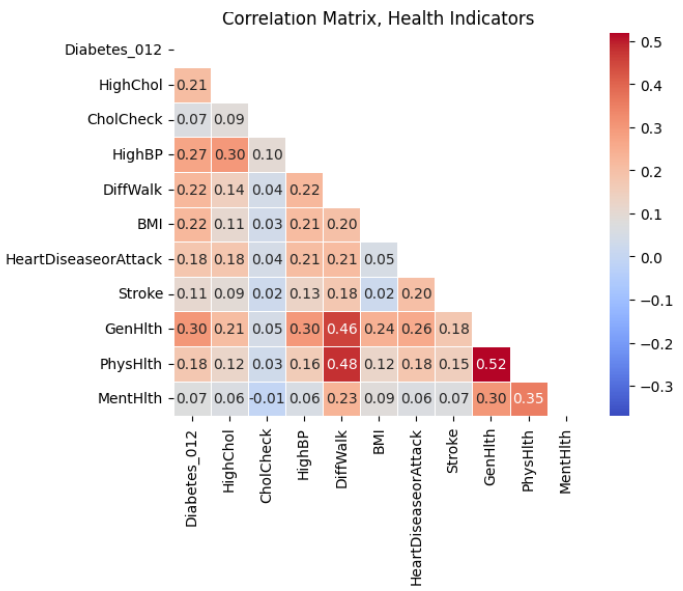

Figure 11 Correlation Matrix for Health features

Regarding health features, most of them are highly correlated, especially with diabetes. For all, the higher the value of the datapoint, the worse the health condition. Therefore, the positive values of the correlation coefficients are cohesive, as several illnesses or health conditions are linked. Only CholCheck is poorly linked with all other health features.

The correlation matrix for individual behaviors depicts a strong correlation between the consumption of fruits and vegetables (0.25), and that practicing a physical activity is made more often by the people eating these items (0.15 and 0.14). They are less likely to have difficulty walking (-0.25) and to be diabetics, which demonstrates that having a 'healthy lifestyle' is slightly more likely to reduce diabetes and favor a good physical condition.

The analysis of these matrices leads to the following conclusions for future data treatment:

- From the first matrix, the consumption of fruits, vegetables and alcohol, the owning of a health care coverage and the financial ability to see a doctor and sex are poorly correlated to diabetes, with a correlation coefficient lower or equal to 0.06 in absolute value.

- From the socio-economic matrix, the education and income levels represent roughly the same people and reduce the risk of diabetes.
- The health matrix shows that all features are linked to each other, and that bad health conditions or diseases favor others.
- The behavior matrix gives low values of correlation coefficients, not only with diabetes, except for one that links poorly correlated features with diabetes.

5.2 Biological Causes of Diabetes

The importance of factors contributing to diabetes varies between type 1 and type 2. For type 1 diabetes, genetic predisposition is the most significant factor, followed by autoimmune reactions that destroy insulin-producing cells in the pancreas. Environmental factors, such as viral infections and diet during early childhood, also play a role, though they are less influential. Age and geography are additional considerations, with type 1 diabetes often diagnosed in younger individuals and incidence varying by region. In contrast, type 2 diabetes is primarily influenced by modifiable lifestyle factors, with obesity and a sedentary lifestyle being the most critical. Family history of type 2 diabetes significantly increases risk, as does poor diet and advancing age. Ethnicity, metabolic syndrome, a history of gestational diabetes, and prediabetes are also important factors. This reflects the global trend where type 2 diabetes is significantly more prevalent due to lifestyle and socio-economic factors. Therefore, the dataset that only contains health and socio-economic data of patients is more relevant to predict type 2 diabetes than type 1 as there is no genetic data in it. However, it is important to note that in the United States, the proportion of diabetes cases is heavily skewed towards type 2 diabetes. Approximately 90% to 95% of all diabetes cases are type 2, while type 1 diabetes accounts for a much smaller percentage (*Diabetes in United States*, 2024). Consequently, the assumption that mostly type 2 diabetics are represented in the dataset is made.

The dataset also contains the data of people suffering from prediabetes which is a condition where an individual's blood sugar levels are higher than normal but not high enough to be diagnosed as diabetes. It is characterized by impaired fasting glucose and/or impaired glucose tolerance, indicating an increased risk of developing type 2 diabetes and associated health complications, such as cardiovascular disease. This condition often goes unnoticed because it is usually asymptomatic, but it serves as a critical warning sign for potential future health issues (Buysschaert & Bergman, 2011).

The literature demonstrates that the most prevalent health factors responsible for type 2 diabetes are BMI and High Blood Pressure. Therefore, only these two health features will remain in the clustering study. When other health features seem biologically irrelevant to predict diabetes, it is considered that features such as General Health or Physical Health are not precise enough to predict it (Banday et al., 2020).

Among behavior factors, Physical Activity and Fruits will remain in the clustering model for sedentarity and poor diet are important factors of type 2 diabetes (*Metabolic syndrome*, 2021). From the socio-economic features, only Income is conserved.

Even if ‘Age’ is not included in sub-analysis of correlation coefficients, it is reasonably correlated with diabetes and poorly correlated with the previous features analyzed.
The features that will be used are summed up below:

- BMI
- High Blood Pressure
- Physical Activity
- Fruits
- Income

5.3 Clustering

The dataset is divided into clusters using the K-means model. The objective is to divide the data with an unsupervised learning method to see if the diabetics categories emerge naturally from the dataset. This is why the number of clusters chosen is equal to three. The results demonstrate that non-diabetics are mostly in cluster 0 and nearly absent from cluster 2. Consequently, non-diabetics are well identified in cluster 0. Pre-Diabetics are equally split between clusters 0 and 1 which means that they are not well identified, but the low number of patients suffering from prediabetes does not strongly influence future studies on the dataset. There are more concerns about Diabetics clustering as they are represented in all clusters. It highlights that the dataset in its original form doesn't enable to clearly identify diabetics.

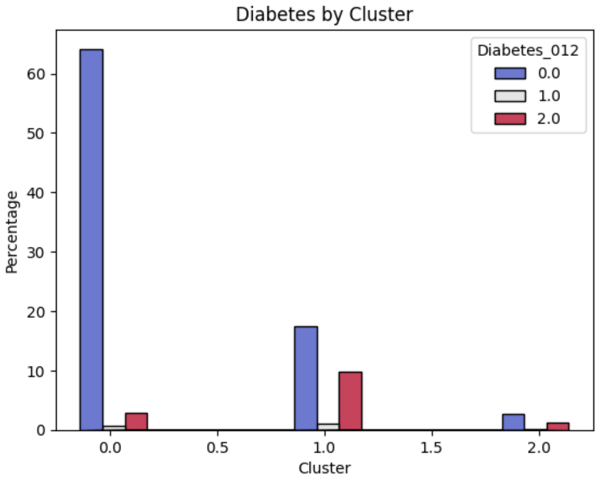

Figure 12 Diabetes by Cluster

As the pre diabetics represent less than 2% of the patients, the choice is made to divide the dataset in only two clusters. The following results are obtained:

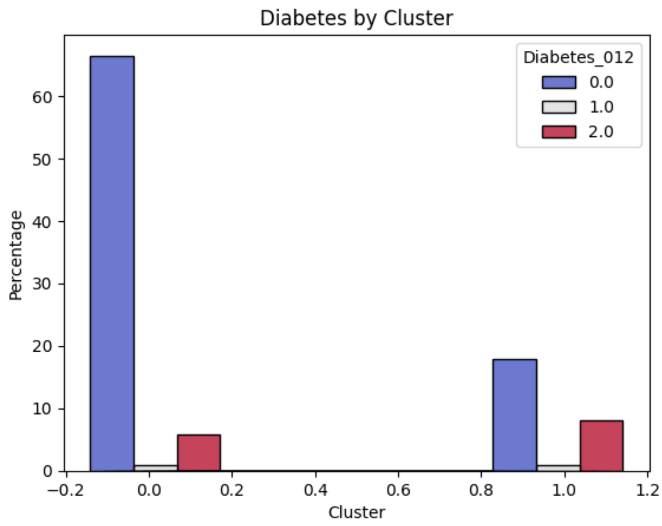

Figure 13 Diabetes by Cluster

When a DataFrame with only the features selected in the Biological Causes for Diabetes section is created (BMI, High Blood Pressure, Physical Activity, Fruits and Income) the two clusters present similar results. Diabetics are mostly concentrated in the second cluster, but at the same time the number of non-diabetics is also higher than in the previous situation. The conclusion drawn from this clustering is that when only the most relevant features for diabetes are considered, the unsupervised learning model K-means doesn't naturally split the diabetics and non-diabetics into two different clusters. It has been tried to implement the DBSCAN method, but the number of clusters created by the algorithm was larger than 100, even when modifying the hyperparameters. When visualizing the repartition of the other features of the dataset, no clear separation appears between the two clusters, meaning the method used doesn't identify naturally two clusters in the dataset studied.

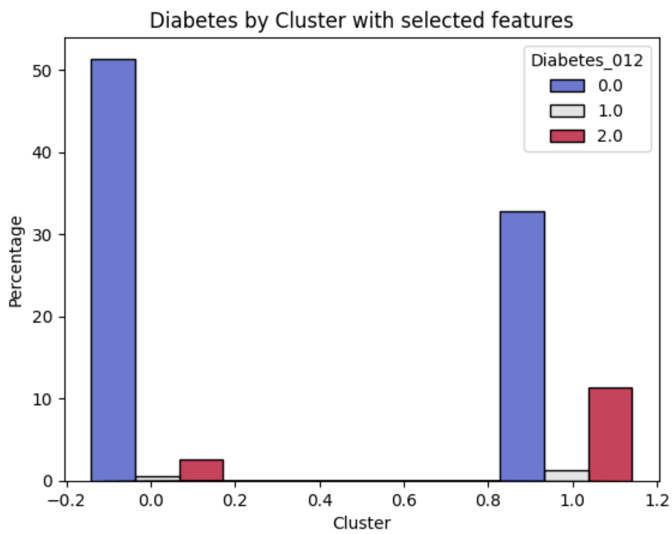

Figure 14 Diabetes by Cluster with selected features

6. Predictive Analysis

Because of the binary classification problem at hand, two methods naturally came to mind as a predictive model. Logistic Regression is commonly used in binary classification problems and allows to fit a sigmoid function to estimate the probability of someone being diabetic for certain input features. On the other hand, Decision Trees work by recursively splitting the data to create rules that classify instances, providing a high level of interpretability. In the Appendix, a comparison between the two different methods is shown. Because of the superior interpretability, easy visualization, and similar accuracy, in this project, Decision Trees are utilized as a predictive model. While individual Decision Trees can suffer from overfitting, particularly when they are allowed to grow deep, a Random Forest mitigates this by averaging the predictions of multiple trees. Each individual decision tree is trained on a random subset of the data. This is called an ensemble method and helps to reduce overfitting and improves generalizability.

Because of the low prevalence of prediabetes (≈1,3% of patients), it has been left out of the predictive analysis. This will help in dealing with class imbalance. To address the class imbalance between non-diabetics and diabetics, the patients without diabetes were randomly under-sampled. Thus, a balanced dataset of roughly 70.000 rows is obtained. This dataset is split into a training and test data set using an 80:20 split. The training data is min-max scaled to a range between 0 and 1. After testing various feature combinations, BPxChol, Age, GenHlth and BMI proved to obtain the best predictive results. BPxChol is the combined effect of HighBP and HighChol. Its value is one if the patient has both high blood pressure and high cholesterol. Otherwise, it is zero. An explanation why this interaction effect was chosen can be found in the appendix. Feature selection and hyperparameter optimization were performed using five-fold cross-validation on the training data to ensure robust and reliable results. The final Random Forest model was trained using the Gini criterion and consists of 20 individual Decision Trees. This number of trees was chosen, because a further increase did not lead to any increase in accuracy. Each of the trees has a max depth of 4, and at most 12 leaf nodes. Limiting the complexity of the individual trees reduces the risk of overfitting and maintains a certain level of interpretability. The general approach in identifying the hyperparameters was to keep the model as simple and interpretable as possible without sacrificing too much performance.

Then, the Random Forest is trained on the whole set of available training data, and its performance is analyzed on the test data. Figure 15 displays the confusion matrix of the model when confronted with the test data. From this, an accuracy of 73% and a F1-score of 73% is calculated. Because the under-sampling of the dataset eliminated the class imbalance, the two metrics are expected to be very similar. In general, the model performs quite well. However, disclosing the genetic disposition of the patient might further improve accuracy. Also, while an accuracy of 73% will give practitioners a good first intuition, it is not high enough to be the only basis for whether a patient is considered diabetic or not. Therefore, a positive classification should only be considered as “has a higher risk of being diabetic” rather than “is diabetic”.

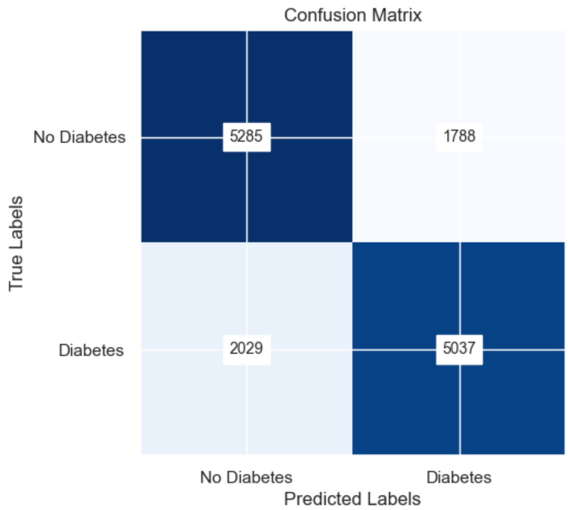

Figure 15 Confusion Matrix of the Random Forest

Figure 16 shows one of the 20 fitted Decision Trees in the Random Forest. The high importance of GenHlth is visible, because it decides the first split at the top node of the decision tree. Mostly, (but certainly not always) the left side of the tree classifies patients as nondiabetics, while the right-hand side of the tree classifies patients as diabetics.

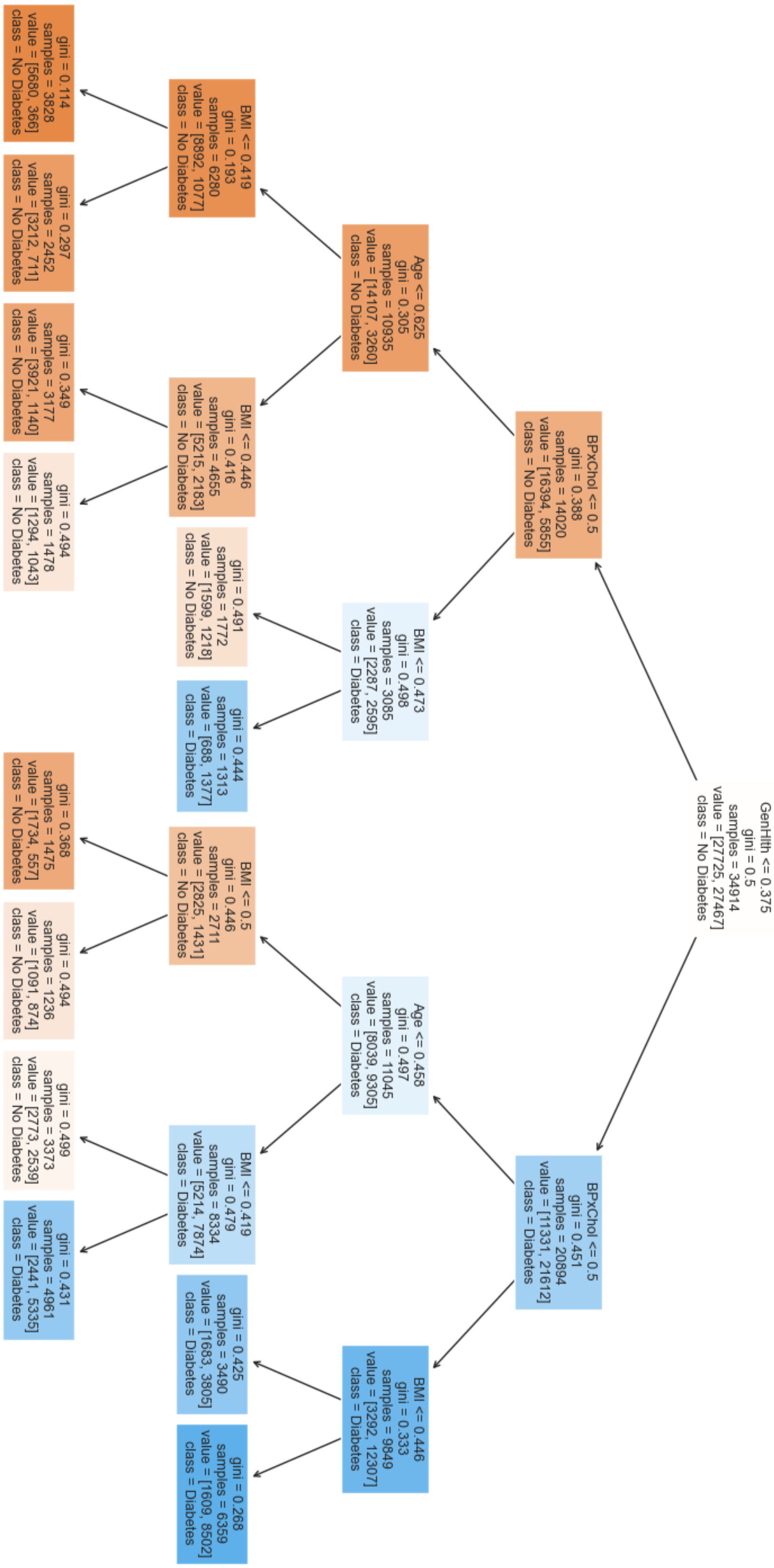

Figure 16 An example of a Decision Tree

7. Prescriptive Analytics

The predictive model delivers a binary classification: each individual is either labeled as diabetic or non-diabetic. Building upon this output, the prescriptive analytics component aims to enable targeted interventions that are data-driven, ethically accepted, and operationally feasible.

Firstly, it is relevant to identify the features that impact the probability of the outcome the most and how they influence the classification result. Shap is a powerful tool to assess and explain the contribution of each feature, for a local or global prediction. Shap can be used to build an expert system, capable of interpreting the predictions and based on the features of the predictive model (*Welcome to the SHAP documentation*, n.d.).

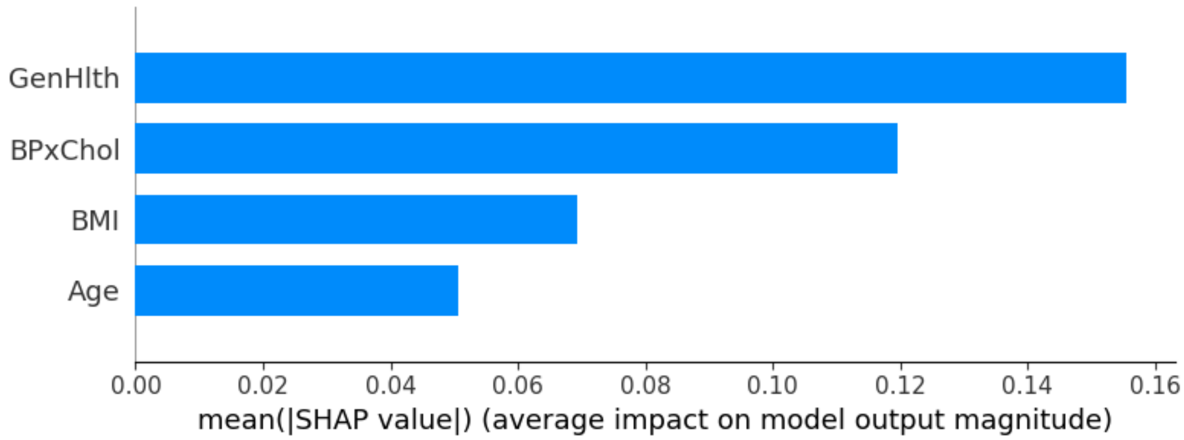

Figure 17 Mean Shap Values for the predictive model

Figure 17 shows the mean SHAP values for the features of the estimated model. It is relevant to know the importance of the features to be aware of the weights to put on the expert system that is being developed. General Health is the feature that impacts the prediction the most for SHAP values, however, BpxChol is very important as well, and a deeper analysis will be conducted with other SHAP plots later.

Even though the model outputs binary predictions, the decisions based on these outputs can be partially automatized using systems which mimic the decision-making abilities of a human expert.

For instance, let's say a patient has the following profile, chosen from the SHAP tool:

- GenHlth = 4
- Age = 11
- BMI = 38
- BPxChol = 1

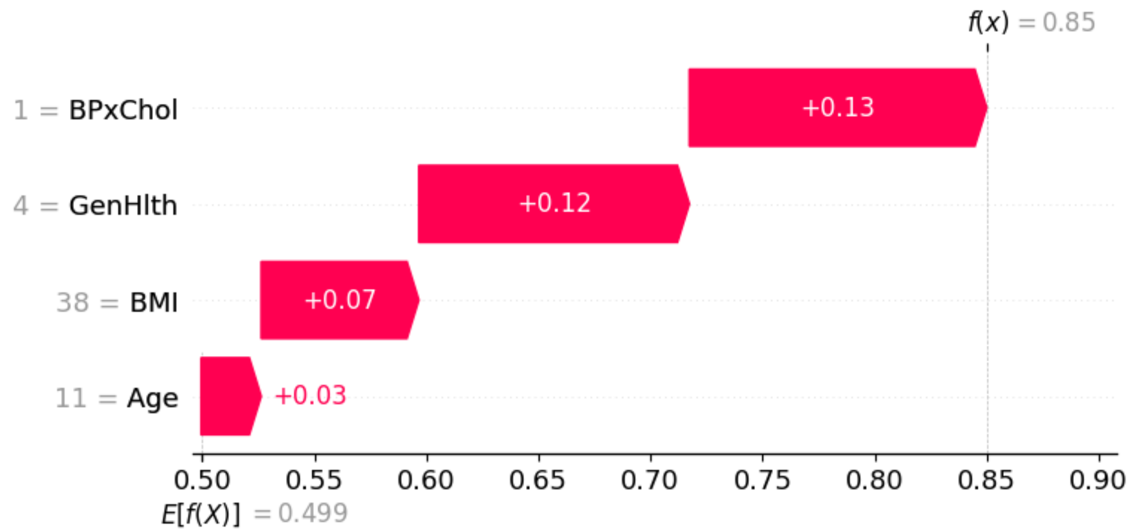

Figure 18 Local Shap values for a single instance

From the SHAP tools, the waterfall plot shows the features responsible for influencing the likelihood of an outcome, providing a Local Explanation. E[f(x)] is the baseline, thus, in this case, the predicted outcome is that the subject is diabetic, since the predicted f(x) is on the right side of the baseline. The plot shows that the strongest feature for that prediction is BPxChol, which increases the general risk of predicting diabetes the most. All the features push the likelihood of the outcome to the classification as a diabetic, which is reasonable, since the data point sorted shows rather negative physical values. The patient, in this case, has High Blood Pressure and High Cholesterol, a poor General Health (=4), a high BMI with respect to the mean population and an age between 70 and 74 years old, which increases its sensibility to illnesses (in this case diabetes). The Local Explanation follows the trend of the mean SHAP values in absolute values, confirming BpxChol and GenHlth at the first positions.

This information may be easily used to know the feature's weight of the expert system for which it is being developed, since it would be reasonable to give priority and privilege to the most sensitive features.

For instance, an expert system would capture the logic using structured rules:

IF (Classification = Diabetic) AND (GenHlth ≥ 4) AND (Age > 60) AND (BPxChol = 1) THEN Recommend: "Schedule diagnostic check-up within 1 week, assign priority level: ***High****"*

And, in case of a less urgent case:

IF (Classification = Diabetic) AND (GenHlth = 2) AND (BMI < 25) THEN Recommend: "Send educational material, routine follow-up in 2 months, assign priority level: ***Low****"*

Furthermore, decisions based on the model's output can be optimized under real-world constraints, based not only on predicted outcomes, but also by considering the real-world limitations of the system implementing those decisions. For instance, let's say the predictive model has classified 6,900 patients as diabetic. However, the healthcare center can only follow up with 1,000 patients per week due to limited staff and testing capacity. Each follow-up includes a consultation and a set of diagnostic tests. To prioritize which 1,000 patients to contact this week to maximize health impact, it is possible to assign a priority score to each diabetic-classified patient based on the importance and on the risk associated to the features:

- +3 points for GenHlth ≥ 4
- +2 points if BPxChol = 1
- +2 points if Age > 60
- +1 point if BMI ≥ 30

Then, we can sort all 6,900 classified patients by this score, select the 1.000 patients with the highest priority scores, schedule their appointments this week and repeat the process next week for the next 1,000, adapting the plan as new data (e.g., cancellations, follow-up results) come in.

However, ethical and fairness implications should be considered, since these systems would treat and give insights for decisions affecting people. For instance, if a person is labeled as diabetic and receives medical advice or is referred to treatment, they have the right to understand why. For this reason, SHAP plots are also useful to explain and interpret the model.

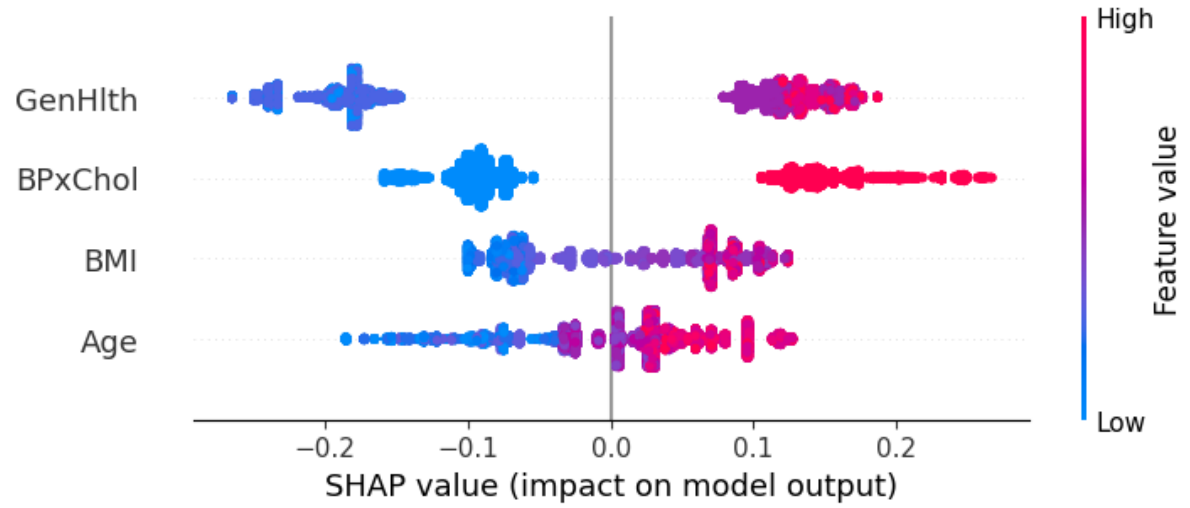

Figure 19 Beeswarm plot of Shap values for all instances

The bee swarm plot shows how the features impact the predictions for each data point, representing their SHAP values with the gradients of color. The impact of having High Blood Pressure and High Cholesterol on the outcome with respect to having only one, or none of them is evident, enhancing the clear threshold between the binary values. Moreover, the broad division of data points is noticeable for GenHlth, which confirms the importance of the attribute among the individual predictions, pushing firmly towards non-diabetic or diabetic outcomes. Moreover, BMI and Age show a more continuous trend of SHAP values. Remarkably, Age has a longer skirt towards the left side (non-diabetic) compared to BMI, which underlines the fact that younger individuals are far less likely to be diabetic than low BMI individuals, even if both, as previously seen, show a lower mean SHAP absolute importance than the other features. However, for some younger predictions, Age still plays a more significant role than BpxChol if classified as 0. This plot can give many insights to explain how the model works and lets the users understand and comprehend how the algorithm is using their data to make predictions. More interpretable scatterplots for each feature are shown in the appendix, which give a clearer picture of the features' values affecting the SHAP values.

In the end, systems must be designed so that accountability remains with human professionals and is not fully delegated to algorithms, for this reason it is fundamental to explain and interpret the model trained.

It is relevant to assess the performance of the model by testing it across different scenarios. For instance, it may be supposed that there's a public health initiative that successfully reduces the average BMI of the population due to better dietary habits and fitness awareness. As a result, the population's BMI is now on average lower than the original dataset used to train the model. For representing the case, the BMI values have been lowered by 10, shifting the mean towards lower values.

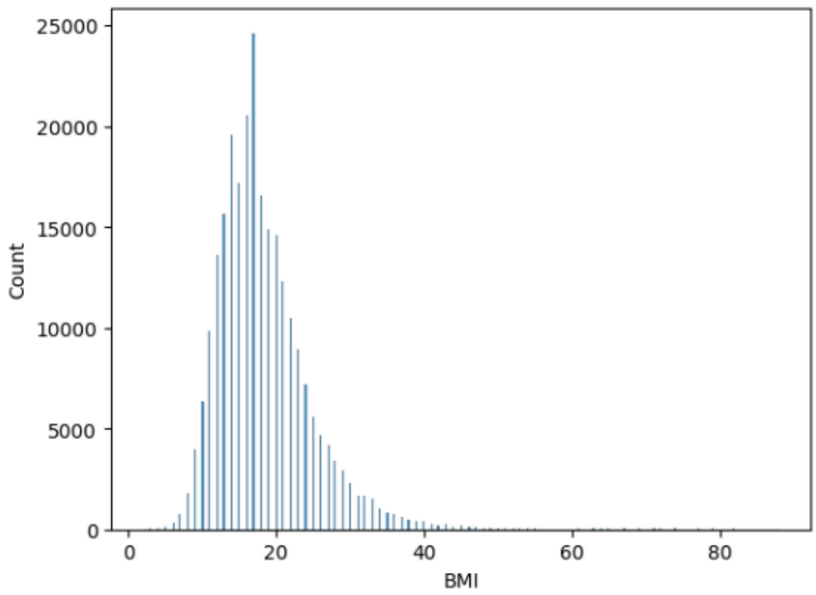

Figure 20 New distribution of BMI after the health intervention

A shift in the BMI distribution has been done by lowering the BMI values in the test set. This will mimic a population that has experienced a decrease in BMI on average due to health improvements. The model's performance showed a decrease in both accuracy (from 73% to 69%) and F1-score (from 73% to 68%).

The drop indicates that the model is sensitive to changes in the distribution of the population. This decline confirms that the model may rely heavily on BMI as a key predictor of diabetes risk, The F1-score drop reinforces the idea that the model's performance degradation is not only due to fewer correct classifications (accuracy drop) but also due to an increased imbalance between false positives and false negatives in the shifted population.

With a decrease in accuracy, there is an increased risk of misclassifying patients. In the case of false positives, patients may undergo unnecessary diagnostic tests, leading to a waste of resources. In the case of false negatives, patients at risk of developing diabetes may not be diagnosed early, which could lead to delayed interventions, worsening their health condition, and increasing healthcare costs in the long run.

8. Fairness and Bias II

Fairness is a critical consideration in data analytics projects, particularly when models influence decision-making in areas like healthcare, hiring, or criminal justice.

8.1 Fairness

Ensuring fairness means addressing biases that can result in discrimination against certain groups. This can be due to historical inequalities, data representation issues or algorithmic decision-making.

One key aspect of fairness is individual vs. group fairness. Individual fairness emphasizes that similar individuals should receive similar predictions, while group fairness focuses on ensuring equitable treatment across demographic groups. However, achieving both at the same time can be challenging, as prioritizing one may compromise the other. For instance, in predictive analytics for diabetes risk, ensuring that high-risk individuals receive proper intervention must be balanced with avoiding disproportionate false positives or false negatives for specific socio-economic groups.

Bias regarding data is another challenge to fairness. Bias can emerge at multiple stages, including data collection, pre-processing, and processing. If the training data disproportionately represents certain groups, the model may learn patterns that reinforce existing disparities. For example, if a diabetes dataset includes fewer records from younger individuals, the model may underpredict risk in this population. Addressing fairness requires proactive bias detection and mitigation techniques, such as reweighting datasets, modifying model objectives (*Fairness and Bias in Machine Learning: Mitigation Strategies*, 2024).

Finally, fairness is a context-dependent concept. Different stakeholders may prioritize different fairness metrics. Ensuring fairness requires transparency, stakeholder involvement, and continuous monitoring to align model outcomes with ethical and legal fairness principles. Additionally, fairness should be treated as an ongoing process that adapts to evolving societal norms and regulatory requirements. By integrating fairness-aware practices, data analytics projects can help mitigate harm and promote equitable outcomes (R, 2023).

8.2 Fairness Analysis

With the labels predicted in the test sample of the predictive analysis, Aequitas is used to assess the fairness of the predictions.

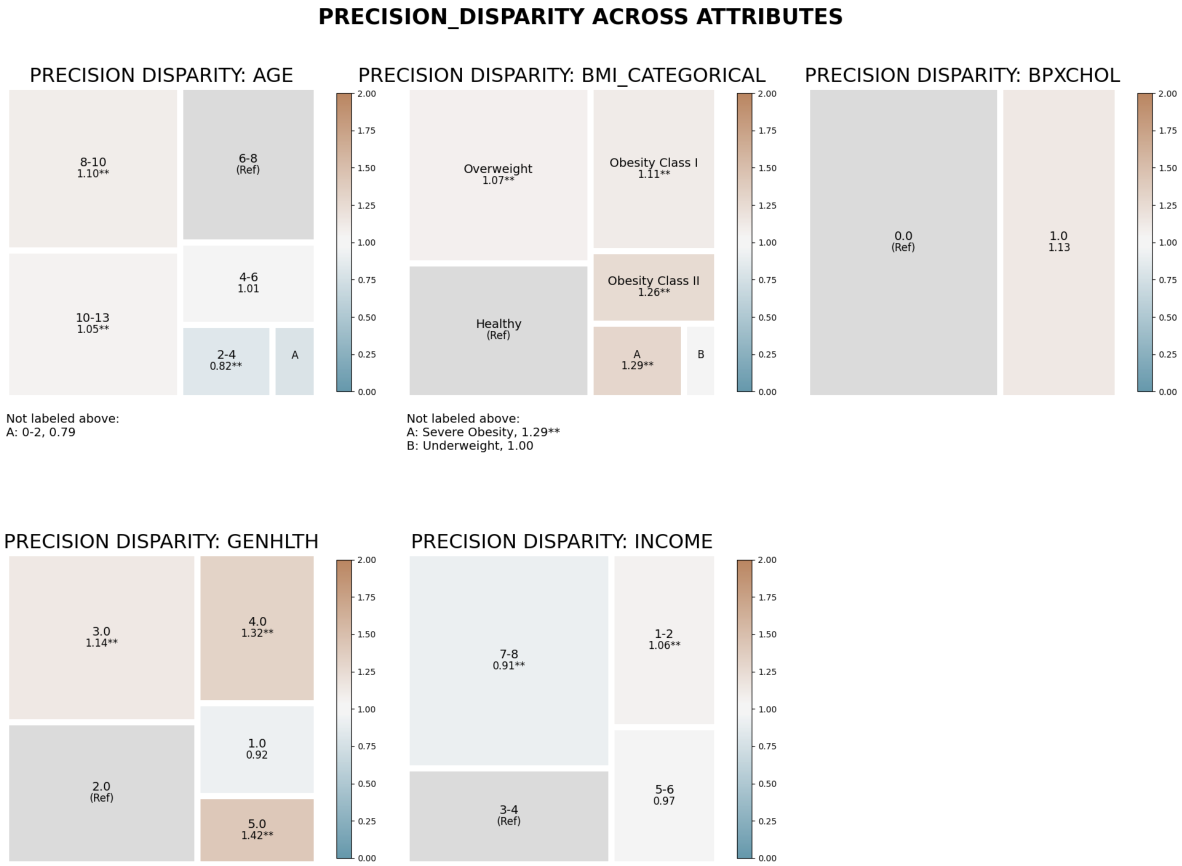

Figure 21 Disparities in precision for different feature values

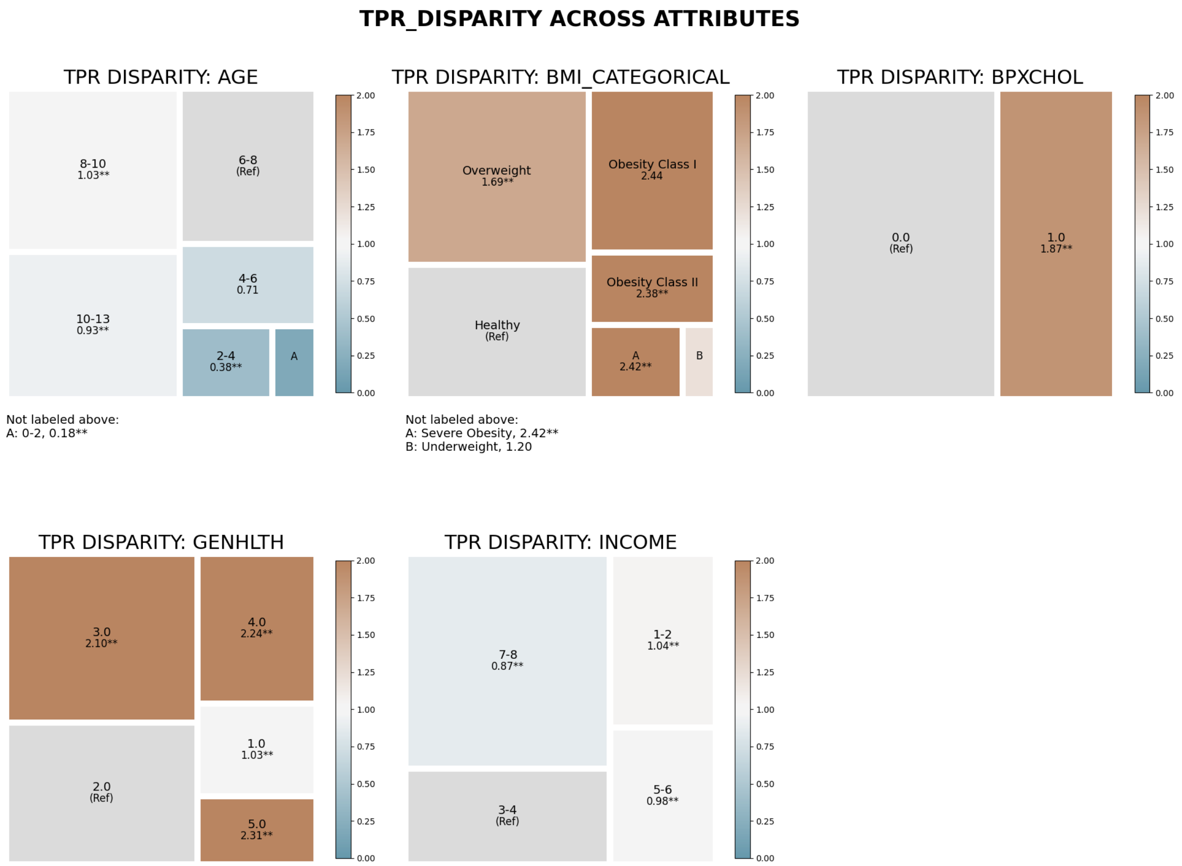

Figure 22 Disparities in True Positive Rate for different feature values

The tree maps above display Precision and True Positive Rate (Sensitivity) disparity values calculated using a predefined group seen as a reference for the calculation. In these visualizations, the reference group is chosen to represent healthy people with an average income and age regarding the patients presented in the dataset. Its characteristics are:

- Age: '6–8' (for age values divided into 13 categories) i.e. people between 45 and 54 years old
- BMI: 'Healthy'
- Blood Pressure x Cholesterol: '0.0'
- General Health: '2.0' out of 5 bins
- Income: '3–4' (for income values divided into 8 categories) i.e. people between 20,000 and 25,000$ revenue a year)

The disparities are calculated based on predefined reference groups. Each square area is proportional to its group size, and the color of squares are based on the disparity magnitude in comparison to the reference group. This is the reason why disparity coefficients are equal to 1 when the feature value is the same as that of the reference group.

It is observed in the precision disparities tree maps (Figure 21) that the model has the same ability to predict exact positive instances for all income, age groups (except for the very young people that are a small part of the patients and for whom the exactness of the model (0.82) is lower and patients suffering from high blood pressure and cholesterol. A pattern can be identified for BMI and General Health categories: the higher the BMI and the worse the general health, the higher the precision. In other words, the model is better at predicting positive instances with few false positives for high BMI and bad general health patients.

Comparing the visualizations of precision and sensitivity (represented by True Positive Rate in Figure 22), the disparities are much more significant for sensitivity based on the reference group. Except for income, every feature presents important disparities. Age has two groups: one for age values superior to 6 that has a higher sensitivity than the patients whose age value is below 6. The BMI, BloodPressure x Cholesterol and general health sensitivity disparities coefficients demonstrate that recall is far higher for people with health issues. An example of this trend is the sensitivity of obese people that is more than twice as high than the one of healthy people, showing that the predictive model used is better at identifying actual positive diabetics without missing them for people that have already health issues.

The combination of these analysis highlights the discriminations created by the model for people that have overall good health, and especially the young ones. As the precision for their group is smaller, they are more likely to be wrongly classified as diabetic. However, the biggest imbalance in the predictions lies in the TPR disparities that are much more important and lead to much more failures in identifying diabetics for this group. This takeaway might be explained by the lack of genetic features that can be predominant to predict diabetes.

8.3 Transparency and Explainability

Transparency about which methods are used, how the input data was pre-processed shows that an organization can defend the choices it has made during the training and testing of a Machine Learning method. It allows an organization to demonstrate its own trust in their decision-making process. Through transparency, stakeholders can build trust in the system they are using and which they depend on, because they know that the organization trusts itself enough to discuss their data analytics process.

Especially on large online platforms, transparency is often lacking. The exact algorithms deciding which videos are recommended next, or why a certain website is ranked above another, are intentionally kept a secret. Often, this is for economic reasons, as a well-performing recommendation algorithm is an important asset and unique feature. Being transparent would dimmish the company's market advantage. Therefore, companies need to find the intricate balance in the tradeoff between keeping their market advantage and gaining the consumer's trust.

While transparency is concerned with what is done, explainability focusses on why a model behaves a certain way. Explainable AI tries to shed light on the decision-making process within a machine learning model. In essence, it is more about providing information about how the model came to a certain outcome. Explainability has become a big point of discussion in recent years, because the models are growing more and more complex. As the complexity of the decision-making process of these large machine learning models is increasing, so is the need for tools that help to look behind the curtain. Libraries such as SHAP and LIME try to illustrate what features are most important in a specific classification.

9. Outputs and Reflections

Building upon the first submission, this report further refines the analysis outputs and reflections on diabetes risk prediction using data analytics. The project has aimed to predict diabetes risk using a dataset containing socio-economic, behavioral, and health-related variables. The dataset contains a large sample size, allowing for detailed statistical analysis but also presenting potential biases due to data collection methods.

The model's performance was evaluated using accuracy, precision, recall, and F1-score. While the predictive model demonstrated reasonable effectiveness, it also highlighted concerns regarding class imbalances in the dataset. Addressing these imbalances through techniques such as reweighting or oversampling could enhance the model's ability to generalize across different population groups.

The project highlighted the ethical and social implications of data analytics, particularly regarding fairness, bias, and representation. The dataset, despite being extensive, may not proportionally represent all demographic groups. Underrepresented populations, such as those with lower digital literacy, non-English speakers, and individuals without stable healthcare access, may have been excluded from the data collection process. This could lead to biased predictions and reinforce existing healthcare disparities.

Additionally, the group positionality was acknowledged in interpreting the data. Coming from European countries with universal healthcare systems, there is a risk of overlooking the structural inequalities in the U.S. healthcare system, where access to medical services is often determined by income and insurance coverage. This awareness is important in assessing the model's applicability and potential misinterpretations.

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11. Appendix

Table 1 Overview of the different variables

Column Name	Variable Type		Explanation
Diabetes_012	Categorical	Ordinal	Diabetes status from the patient
HighBP	Categorical	Ordinal	Adults who have been told they have high blood pressure by a doctor, nurse, or other health professional
HighChol	Categorical	Ordinal	Have you EVER been told by a doctor, nurse or other health professional that your blood cholesterol is high
CholCheck	Categorical	Binary	Cholesterol check within past five years
BMI	Quantitative	Continuos	Body Mass Index (BMI)
Smoker	Categorical	Ordinal	Have you smoked at least 100 cigarettes in your entire life?
Stroke	Categorical	Binary	(Ever told) you had a stroke.
HeartDiseaseorAttack	Categorical	Binary	Respondents that have ever reported having coronary heart disease (CHD) or myocardial infarction (MI)
PhysActivity	Categorical	Ordinal	Adults who reported doing physical activity or exercise during the past 30 days other than their regular job
Fruits	Categorical	Ordinal	Consume Fruit 1 or more times per day
Veggies	Categorical	Ordinal	Consume Vegetables 1 or more times per day

HvyAlcoholConsump	Categorical	Ordinal	Heavy drinkers (adult men having more than 14 drinks per week and adult women having more than 7 drinks per week)
AnyHealthcare	Categorical	Binary	Do you have any kind of health care coverage, including health insurance, prepaid plans such as HMOs, or government plans such as Medicare, or Indian Health Service?
NoDocbcCost	Categorical	Binary	Was there a time in the past 12 months when you needed to see a doctor but could not because of cost?
GenHlth	Categorical	Ordinal	Would you say that in general your health is
MentHlth	Quantitative	Discrete	Now thinking about your mental health, which includes stress, depression, and problems with emotions, for how many days during the past 30 days was your mental health not good?
PhysHlth	Quantitative	Discrete	Now thinking about your physical health, which includes physical illness and injury, for how many days during the past 30 days was your physical health not good?
DiffWalk	Categorical	Binary	Do you have serious difficulty walking or climbing stairs?
Sex	Categorical	Nominal	Indicate sex of respondent.
Age	Categorical	Ordinal	14 age levels. Fourteen-level age category
Education	Categorical	Ordinal	What is the highest grade or year of school you completed?
Income	Categorical	Ordinal	Is your annual household income from all sources: (If respondent refuses at any income level, code "Refused.")

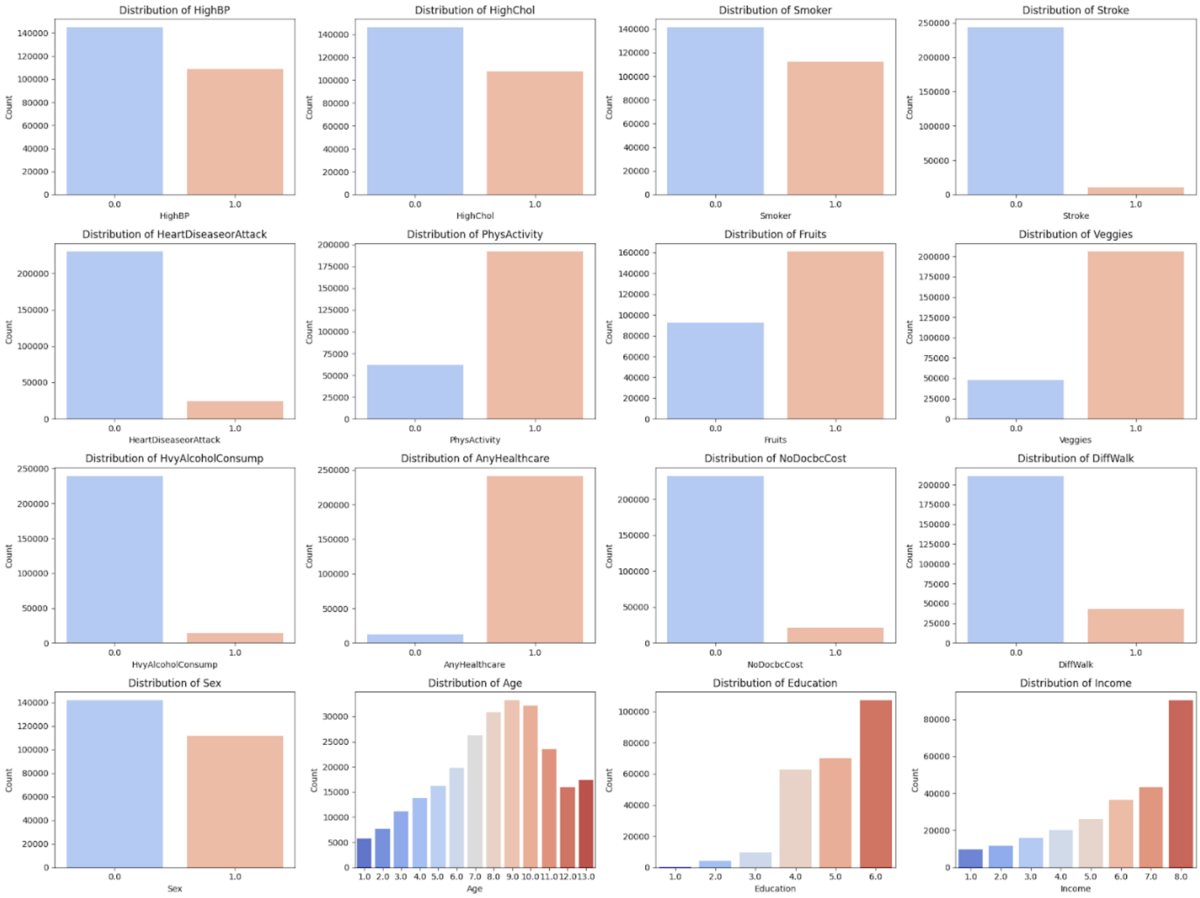

Figure 23 Overview of the Distribution of all available features

Picking The Predictive Model

This subsection makes a comparison between Decision Trees and Logistic Regression in order to find out which method is more suited for the modeling problem at hand. Because the dependent variable is binary, Logistic Regression is a promising method. However, because of the mostly categorical or binary nature of the independent features, a Decision Tree could also be a good method. In the following a, decision tree, and a Logistic Regression are trained. This allows for a comparison of which methodology is more practical for this dataset and use case.
In both the Logistic Regression and the Decision Tree, the respective set of features was normalized before fitting the model. Furthermore, the features for both methods were selected automatically by scikits automatic feature selection. It selects the most important features in order to increase model performance. Because pre-diabetes is only present in roughly 1.8% of patients it was left out of this regression to avoid high class imbalance. The performance of the model is measured by its overall accuracy and F1-score. Because diabetic patients are less common than non-diabetic patients, accuracy alone is not a sufficient metric. The F1-Score is the harmonic mean between accuracy and recall and less vulnerable to class imbalance. To ensure robustness in the results, cross validation with five folds is utilized. The results can be seen in the following two tables. The confusion matrices are obtained by using fivefold cross validation prediction.

Comparing the differences between the two methods used, their performance metrics are very similar. Because of that, the decision about which method is more suited for the needs of this project is not guided by performance, but more by general characteristics of the two different methods. While the Logistic regression allows clear interpretation of features because of tractable coefficients, the Decision Tree provides accessible decision boundaries, and its functionality is also easily visualizable. This makes the Decision Tree an intuitive method to use for predictive analytics.

Table 2

Overview	**Logistic Regression**		**Decision Tree**	
HighChol	0.3			
CholCheck	0.24			
HighBP	0.39		X	
BMI	0.5		X	
GenHlth	0.68		X	
Age	0.5		X	
HvyAlcohol Consumption	- 0.17			
Accuracy	73.11% (sd = 0.0)		71.4% (sd = 0.0)	
F1	0.63 (sd = 0.0)		0.62 (sd = 0.0)	
Confusion Matrix	*no diabetes*	*diabetes*	*no diabetes*	*diabetes*
no diabetes	154.940	58.763	149.330	64.373
diabetes	8.206	27.140	8.354	26.992

The Interaction Effect between HighBP and HighChol

High Blood Pressure and High Cholesterol can both be indicators of an unhealthy lifestyle. However both these symptoms can also stem from a genetic disposition. While still possible when combining the two features, the interaction effect between the two increases the likelihood that high cholesterol and high blood pressure are due to unhealthy practices instead of genetic disposition. Furthermore, Figure 24 shows a scatterplot of patients with respect to High Blood Pressure and High Cholesterol. It is noticeable, that especially for patients with both High Blood Pressure and High Cholesterol, the number of diabetics in that group is large.

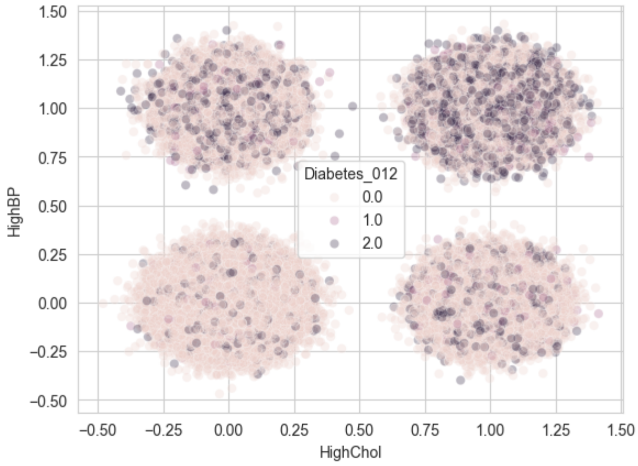

Figure 24 Scatterplot of patients

Scatterplots of The Features and SHAP Values

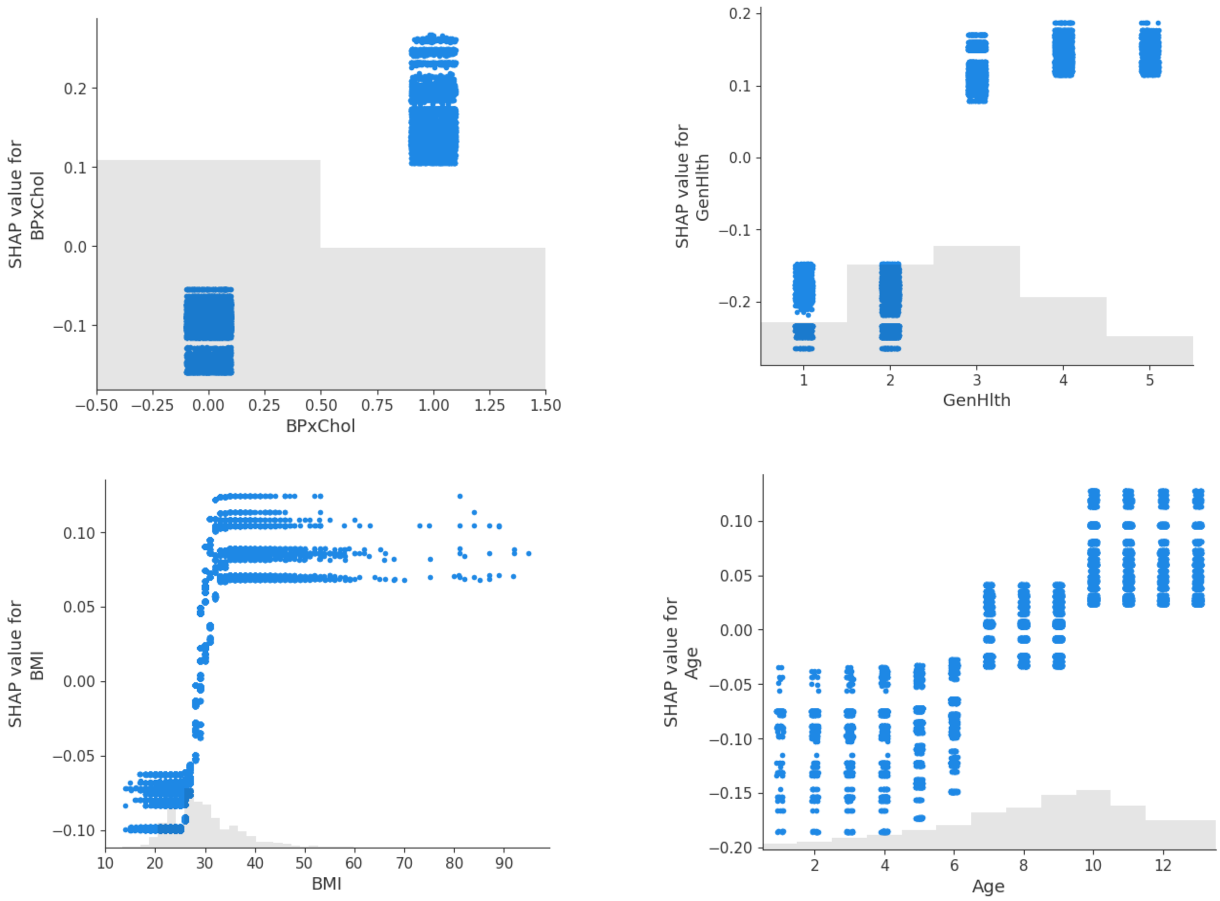

Figure 25 Scatterplots for each Feature and the respective SHAP Values